Sultan Mahamud Opu

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Objective

To build an academic career as a Lecturer in Computer Science and Engineering by contributing to quality teaching, student supervision, and research. My interests include Machine Learning, Explainable AI, Artificial Intelligence, and Software Development, supported by research publications, academic projects, and hands-on development experience. I aim to foster students' technical and analytical abilities while pursuing impactful research and contributing meaningfully to the academic community.

Education

- **Hajee Mohammad Danesh Science and Technology University** — *March, 2022 – August, 2026*
  B.Sc. in Computer Science and Engineering — Dinajpur, Bangladesh
  - CGPA : 3.69/4.00 (Ranked 5th in Batch)
- **Dr. Mahbubur Rahman Mollah College** — *2018 – 2020*
  Higher Secondary Certificate (Science) — Dhaka, Bangladesh
  - GPA: 4.67/5.00
- **Bandura Holy Cross High School & College** — *2016 – 2018*
  Secondary School Certificate (Science) — Dhaka, Bangladesh
  - GPA: 4.89/5.00

Experience

- **AU-Ltd** — *March, 2026 – Present*
  Mobile Application Developer
  - Developed FitWell-Flow from scratch to Play Store and App Store release, implementing authentication, user profile updates, REST API integration, activity tracking, clock-in/out, and leave management.
  - Implemented authentication features for the Eshogol mobile application.

Research Work

Publications: **C=Conference, U=Under Review, E=Extended Abstract, D=Dataset Contribution**

[U.1] Najma Akter Lopa, **Sultan Mahamud Opu**, Md. Tanvir Ahmmed Shuvo and Pankaj Bhowmik, *"Towards Responsible Clinical AI: An XAI-Driven Framework for Cardiovascular Disease Risk Assessment with Clinical Decision Support"*, 29th International Conference on Computer and Information Technology (ICCIT 2026), IEEE

[C.1] Sourav Roy, Najma Akter Lopa, **Sultan Mahamud Opu**, Shafiqul Islam Sazu, Md. Ridwanur Rahman Shuvo, Raghubir Chaudhary, Ashis Kumar Mandal and Pankaj Bhowmik, *"Explainable Machine Learning Framework for Early Detection of Depression in University Students"*, 2026 IEEE International Conference on Engineering and Frontier Technologies (ICEFronT 2026), IEEE (Accepted)

[D.1] Najma Akter Lopa, **Sultan Mahamud Opu**, Sourav Roy, Mst Ifat Zahan Soma and Pankaj Bhowmik (2026), *"A Cross-Sectional Dataset of Sleep Disorder Indicators Among Bangladeshi University Students"*, **[Thesis-based Primary Dataset]**, *Mendeley Data*, V1. DOI: 10.17632/9djgjvtj8d.1.

[C.2] Najma Akter Lopa, **Sultan Mahamud Opu**, Sourav Roy, Mst. Ifat Zahan Soma, and Pankaj Bhowmik (2026). *"AI-Driven Analysis of Sleep Disorders in Bangladeshi University Students Using Clinical and Behavioral Insights"*. In 5th IEEE International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE 2026), **[Thesis-based Conference]**, DOI: 10.1109/ICECTE69292.2026.11429299

[E.1] Sourav Roy, Najma Akter Lopa, **Sultan Mahamud Opu**, Shafiqul Islam Sazu, Md. Ridwanur Rahman Shuvo, Raghubir Chaudhary, Ashis Kumar Mandal and Pankaj Bhowmik, *"Enhancing Early Detection of Depression in Academic Settings: An XAI-Guided Machine Learning Approach for Bangladeshi University Students"*. In Scholars' Summit 2025, HSTU Research Society. **[Accepted]**, Certificate

[E.2] Najma Akter Lopa, **Sultan Mahamud Opu**, Sourav Roy and Pankaj Bhowmik,*"Towards an AI-Driven*

Framework for Effective Sleep Disorder Prediction Analyzing Clinical and Behavioral Insights", RUEC 1st International Research Conference (2025). **[Accepted]**, Certificate

Thesis: **D=Defense Completed, W=Working on it**

[D.1] **Sultan Mahamud Opu**, Najma Akter Lopa, Sourav Roy (2026). *"Sleep Disorder Risk Assessment and AI-Based Clinical Decision Support Among University Students"*. In **Level 4 Semester II** of B.Sc. (Engineering) in Computer Science and Engineering, supervised by Pankaj Bhowmik and Dr. Masud Ibn Afjal.

[D.2] **Sultan Mahamud Opu**, Najma Akter Lopa, Sourav Roy (2025). *"A Comparative Analysis of Gender Disparity in Programming Skills Using Educational Data Mining"*. In **Level 4 Semester I** of B.Sc. (Engineering) in Computer Science and Engineering, supervised by Pankaj Bhowmik and Dr. Masud Ibn Afjal.

PROJECTS

• StudyForge AI *2026*
Tools: Python, LangChain, Gemma, ElevenLabs, Flask Live Demo

- Developed an AI-powered learning platform with AI Tutor, Study Planner, Smart Quiz, and PDF Reader.
- Implemented personalized study plans, interactive quizzes with instant scoring, and PDF-grounded chatbot responses.
- Integrated Bangla-English support, text-to-speech, and PDF downloading.

• HSTU Student Complaint Management System *June 2026 – August 2026*
Tools: HTML, CSS, JavaScript, Backend, MySQL Live Demo

- Developed a web-based complaint management system with unique complaint ID tracking.
- Implemented complaint status tracking, updates, and student feedback.
- Built an admin panel for complaint review, status management, and monitoring.

• Personal Local Cloud Storage and File Sharing System *May 2026 – July 2026*
Tools: Python, FastAPI, MySQL, HTML, CSS, JavaScript, Ngrok Personal Cloud

- Developed a local cloud platform for file storage, organization, and sharing.
- Implemented QR/IP-based access, folder management, dashboards, and file recovery.
- Added encrypted storage and external file sharing using Ngrok.

• Diabetes Prediction System *2026*
Tools: Python, Scikit-learn, SHAP, Gradio Live Demo

- Built a diabetes prediction model on 2,019 records with 15 features, achieving 95.67% accuracy.
- Applied preprocessing, scaling, normalization, and Bayesian hyperparameter optimization.
- Evaluated robustness using ROC-AUC, stratified 5-fold cross-validation, and stress testing; analyzed feature importance with SHAP.

• Tic-Tac-Toe AI Game *2026*
Tools: HTML, CSS, JavaScript, Minimax Live Demo

- Developed an interactive Tic-Tac-Toe game with three AI difficulty levels.
- Implemented Minimax with alpha-beta pruning and deployed via GitHub Pages.

• HSTU_S&T_Hub: Real-Time Communication System *January 2025 – June 2025*
Tools: Flutter, Android, IOS, Web, Supabase, Firebase HSTU_S&T_Hub

- Developed a platform for teachers and Class Representatives to manage schedules, rooms, and academic communication.
- Integrated real-time messaging, role-based access, and an AI chatbot using Firebase and Supabase.

LEADERSHIP & ACTIVITIES SECTION

• IEEE Computer Society HSTU Student Branch Chapter *March, 2026 – September, 2026*
Treasurer

• IEEE Computer Society HSTU Student Branch Chapter *January, 2025 – September, 2026*
Publication Secretary

• Programmers Arena (HSTU) *September, 2025 – Present*
Office Secretary

• Programmers Arena (HSTU) *January, 2025 – September, 2025*

Publication and IT Secretary

- **Programmers Arena (HSTU)** *February, 2024 – January, 2025*
  Executive Committee Member
- **HSTU Research Society (HSTURS)** *2023 – 2024*
  Executive Committee Member
- **CSE Club of HSTU** *January, 2022 – January, 2023*
  IT Secretary
- **Team Matrix – Ethical Hackers** *January, 2022 – 2026*
  Campus Ambassador

Achievements and Awards

- **Honorable Dean's Award** *2026*
  - Received the Honorable Dean's Award from the Faculty of Computer Science and Engineering, HSTU.
- **Winner – Hacktoberfest Hack Day Dinajpur 2026** *October, 2026*
  - Best Open-Source AI Project
  - Best Project Built with ElevenLabs
- **Runner-up – Scholars' Summit 2025** *November, 2025*
  - *Enhancing Early Detection of Depression in Academic Settings: An XAI-Guided Machine Learning Approach for Bangladeshi University Students*, Segment A (Science & Engineering), HSTU Research Society.
- **4th Place – Rangpur Divisional Project Showcasing** *June, 2025*
  - *HSTU_T&S_HUB: A Real-Time Communication System.*
- **Participant – Learnathon (Organized by Brain Station 23)** *January, 2025*
- **7th Place – Rangpur Divisional Collegiate Programming Contest (BRUR)** *October, 2024*

Skills

- **Languages:** Strong reading, writing, and speaking competencies in Bangla and English.
- **Programming Languages:** C, C++, Java, Python, Dart.
- **Web Technologies:** HTML5, CSS3, FastAPI.
- **Frameworks:** Flutter, JavaFX, Streamlit.
- **Database Systems:** MySQL, Firebase, Supabase.
- **Data Science & Machine Learning:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch, SHAP, LIME.
- **Machine Learning:** Classification, Regression, Ensemble Learning, Feature Engineering, Feature Selection, Hyperparameter Optimization, Cross-Validation.
- **Specialized Area:** Explainable AI (XAI), Stacking Ensemble, Bayesian Optimization, Statistical Analysis, ML Model Evaluation.
- **Deep Learning & NLP:** Neural Networks, CNN, RNN, LSTM, Transformers, Natural Language Processing, Text Classification, Text Preprocessing.

References

1. **Pankaj Bhowmik**
   Assistant Professor, Department of Computer Science and Engineering
   Hajee Mohammad Danesh Science & Technology University, Dinajpur
   Email: pankaj.cshstu@gmail.com
   Phone: +8801791848439
   Relationship: [e.g., Thesis Supervisor, Research Advisor & Research Collaborator etc.]

2. **Dr. Masud Ibn Afjal**
   Professor, Department of Computer Science and Engineering
   Hajee Mohammad Danesh Science & Technology University, Dinajpur
   Email: masud@hstu.ac.bd
   Phone: +8801737049633
   Relationship: [e.g., Research Advisor, Thesis Co-Supervisor, etc.]

DECLARATION

I hereby declare that all the information provided above is true and correct to the best of my knowledge and belief.